

NAGARAJA MARKAPURAM

Principal Frontend Architect

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SUMMARY

Principal Frontend Architect specializing in large-scale UI platforms across Angular and React ecosystems. Proven track record of enabling engineering organizations at scale through micro-frontend architectures, shared design systems, and robust frontend infrastructure.

EXPERIENCE

Principal Frontend Architect / Staff II UI Engineer, VMware Cloud Foundation Division

VMware by Broadcom

07/2011 - Present Bengaluru, India

VMware Cloud Foundation (enterprise cloud platform combining compute, networking, storage, and management services)

- Led Clarity Design System development, creating widely adopted web components. Improved UI consistency; accelerated development cycles.
- Developed a frontend platform for Angular and React using Micro-Frontends and modular architecture. Enhanced scalability and enabled independent team delivery.
- Established frontend architecture governance and mentored 25+ engineers globally, driving adoption of best practices, improving code quality, and reducing production defects.
- Led and mentored a distributed frontend team (30+ including contract workforce), driving delivery, technical direction, and execution across large-scale UI platform initiatives.
- Drove performance optimization initiatives, improving platform load times and responsiveness through modular design, efficient bundling strategies, and frontend architecture enhancements.
- Hands-on development across Customer Connect (formerly My VMware) and VCF Business Service Console platforms, contributing to critical features and resolving complex production issues.
- Developed proof-of-concept applications using React and modern tooling to evaluate architecture patterns, enabling informed decisions on cross-framework strategy and adoption.
- Reviewed and guided React-based implementations across teams, ensuring adherence to architecture standards, performance best practices, and scalable component design patterns.
- Led AI-driven initiatives leveraging Gemini and Claude APIs to automate frontend patterns, boosting developer productivity and reducing UI component development time.

Team Leader, Frontend Engineering

United Online Software Development Pvt Ltd

08/2005 - 07/2011 Hyderabad, India

United Online Software is a leading provider of consumer Internet products and services

- Led UI engineering team delivering 'Email on the Web' platform.
- Developed reusable JavaScript component framework, improving development efficiency and consistency across applications.
- Pioneered video mail and chat optimized for low-bandwidth environments.
- Optimized the web email application for the legacy browsers (IE6+, Firefox, Opera, etc...)
- Improved application performance by 30% by automating tree-shaking, optimizing stylesheets, and introducing a dedicated static asset server.

CORE ARCHITECTURAL COMPETENCIES

- Enterprise Design Systems * Micro-Frontend Architecture * UI Platform Engineering * Developer Experience Platforms * Frontend Observability * Performance Engineering * Frontend Governance

KEY ACHIEVEMENTS

- Frontend Standards Definition**
Defined Division-wide frontend standards for Reusable components consumption, accessibility, and performance

- VMware Innovation Hub Selection**
Selected for VMware Innovation Hub for UI platform innovation

- 'At Our Best' Recognition**
Recipient of 'At Our Best' recognition and Corporate Mobile Hackathon winner

SKILLS

Languages

TypeScript JavaScript (ES6+) HTML5

CSS3

Frontend

Angular React RxJS

Web Components (Lit) Tailwind CSS

State Management

NgRx Redux Context API

Build, Testing & Tooling

Vite Parcel Babel Webpack

Husky Rollup Node.js Storybook

Jest React Testing Library Cypress

Vitest Jasmine Protractor

SonarQube

Platform & Infrastructure

Docker Kubernetes GitHub CI/CD

Cloudflare Akamai

Observability & APIs

Dynatrace ThousandEyes

Apache Superset GraphQL REST APIs

AI Tools

Cursor Claude ChatGPT API

Gemini

Architecture

Micro-Frontends Module Federation

Nx Monorepos

Component-Driven Development

EDUCATION

Bachelor of Technology in Electronics and Computer Stream

Acharya Nagarjuna University

- 82% (University 3rd Rank)